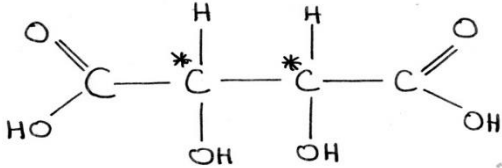


Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
8.	(a)	(i)		200cm ³ at 100°C contains 8.0 g 200cm ³ at 20°C contains 0.32 g (1) amount precipitated = 8.0 – 0.32 = 7.68 / 7.7 g (1) answer must be to two or three significant figures		2		1		2
		(ii)	I	loss of CO ₂ / removal of a carboxyl (or COO) group	1				1	1
			II	award (1) for balanced equation award (1) for structure of the <i>E</i> form of the diene $\text{CH}_3(\text{CH}_2)_4\text{COO}^- \text{Na}^+ + \text{NaOH} \rightarrow \text{CH}_3(\text{CH}_2)_3\text{CH}=\text{CHCOOH} + \text{Na}_2\text{CO}_3$			2			2
			III	0.4	1					1
	(b)	(i)		 award (1) for either chiral centre identified	1					1